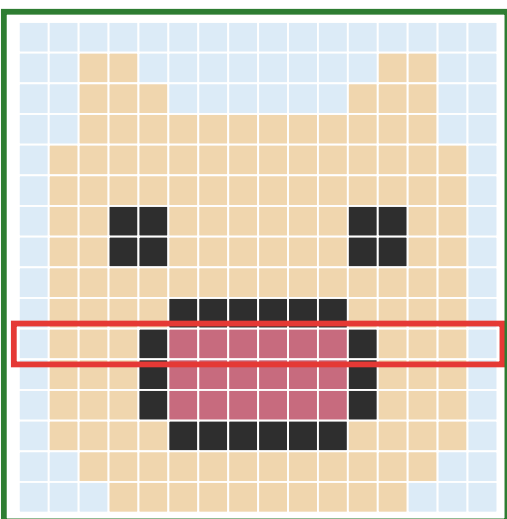


How an image maps onto vector hardware: 1D processor vs my 2D processor

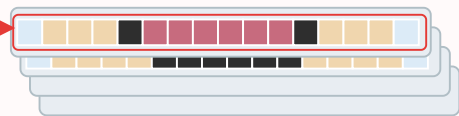
Input image from camera (e.g. 16×16)



rows → memory

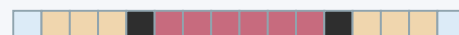
Stock 1D vector machine

Vector register stacks: 1 vector register = 1 row



load 1 row at a time

1D vector processor

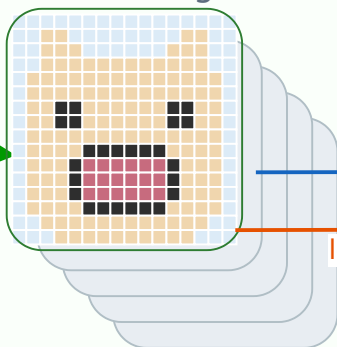


Process one image row at a time

rows → memory (stream each row to fill up the vector register plane)

2D vecotor machine

2D Vector register stack:
1 vector register = 1 data plane



load rows (H)

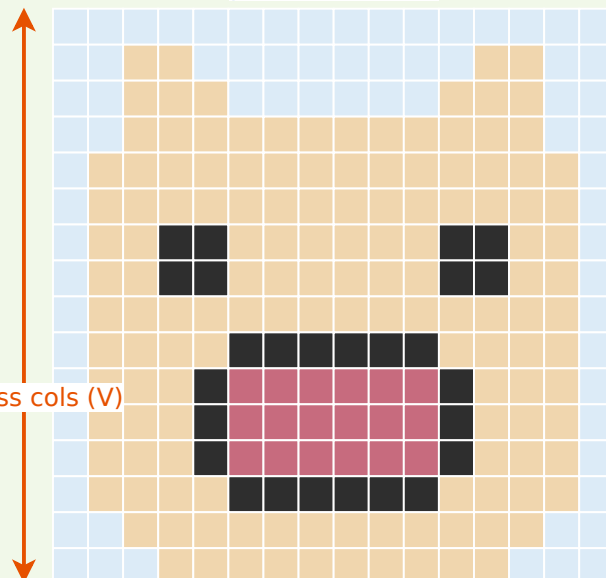
load columns (V)

process cols (V)

No round trip back to memory to
switch between transpose with
horizontal & vertical processing

2D processor array (each row is one 1D processor)

process rows (H)



Process multiple image row at a time